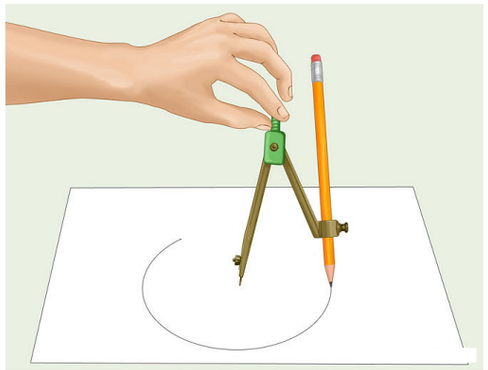
1.1.2 Drawing a circle using compass



A compass is a geometrical instrument which has two arms, one is a needle and the other one is a pencil holder. We use compass to draw circles.

EXAMPLE

Draw a circle of radius 5cm using a compass.



Step 1 Take a compass and fix the pencil in it.

Step 2 Set the distance between two arms of the compass as 5cm using a scale.

Step 3 Fix firmly the needle of the compass on a point which is marked on the paper.

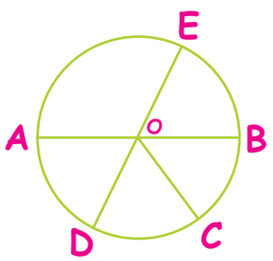
Step 4 Move the pencil around it in any direction till you reach the starting point.

Exercise 1.3

Draw circles for the following measurements.

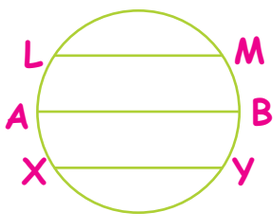
a. 6 cm b. 5.5cm c. 8cm d. 6.8cm e. 8.6cm

1.1.3 Identifying center, radius and diameter of a circle



Circle is perfectly round in shape. It has no **sides** and no **diagonals**. '**O**' is the center of the circle. The distance from the center to each of these points A, B, C, D,E is the **radius** of the circle. Radius is equal in length (OA=OB=OC=OD=OE).

All the radii are equal in length in a circle.



A Line segment AB passes through the center of the circle O .AB is the diameter of the circle. The line segments XY and LM are the **Chords** (XY, LM, AB). The longest chord of a circle is the **diameter** (AB). The radius is always half of the diameter.

Diameter is the longest chord.

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